USN | | | | | | | | | | [22CSE274]

FIFTH SEMESTER B.Tech EXAMINATION, DECEMBER - 2024

FUNDAMENTALS OF ARTIFICIAL INTELLIGENCE

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART-A		**5 x2=10**	
1.	**a.**	Define the term AI.	L2	CO2	**02**
	b.	List down the characteristics of intelligent agent.	L2	CO3	**02**
	c.	What is heuristic search strategy?	L3	CO4	**02**
	d.	With an example, show objects, properties functions and relations.	L2	CO5	**02**
	e.	What are the different types of planning?	L2	CO5	**02**
		PART-B		**6 X 5=30**	
2.	**a.**	What are the risks and benefits of artificial intelligence?	L2	CO1	**05**
		OR			
	b.	Discuss Types of knowledge-based systems.	L2	CO1	**05**
3.	**a.**	Give PEAS description for different agent types.	L2	CO2	**05**
		OR			
	b.	Discuss on the Structure of the Intelligent AI.	L2	CO2	**05**
4.	**a.**	Differentiate toy problems and real world.	L2	CO4	**05**
		OR			
	b.	Differentiate Uninformed Search (Blind search) and Informed Search (Heuristic Search) strategies.	L3	CO4	**05**
5.	**a.**	Write various Knowledge Representation issues.	L3	CO4	**05**
		OR			
	b.	Differentiate Forward chaining and backward chaining.	L3	CO4	**05**
6.	**a.**	Brief on classical planning.	L3	CO5	**05**
		OR			
	b.	Distinguish between Deterministic and Non-deterministic Algorithms.	L3	CO5	**05**
7.	**a.**	What is planning in AI? Describe the role of planning in AI.	L2	CO6	**05**
		OR			
	b.	Demonstrate the components of a planning system.	L3	CO6	**05**
		PART-C		**6 X 10=60**	
8.	**a.**	Explain Evolution of Artificial Intelligence: From Perceptron to Big Data.	L2	CO1	**10**
		OR			
	b.	Describe the importance of PEAS in AI. Consider a Hospital management System and give it's PEAS description.	L2	CO3	**10**
9.	**a.**	Sketch the structure of agent with example.	L2	CO3	**10**
		OR			
	b.	Explain Agent Types in AI.	L2	CO3	**10**

JAIN (Deemed-to-be University) Learning Resource Centre Jain Global Campus Ramanagara-562112

10.	**a.**	Three missionaries and three cannibals are on one side of a river, along with a boat that can hold one or two people. The goal is to find a way to transport everyone to the other side of the river without ever leaving a group of missionaries in one place outnumbered by the cannibals in that place. - Formalize the problem into state space search - Design algorithm with suitable data structure.	L3	CO4	**10**
		OR			
	b.	Given a 3×3 board with 8 tiles (each numbered from 1 to 8) and one empty space, the objective is to place the numbers to match the final configuration using the empty space. You can slide four adjacent tiles (left, right, above, and below) into the empty space. Initial state: Final state: - Design A* algorithm to solve the above problem. The number of misplaced tiles is used to calculate the heuristic value (h-score) in the A* algorithm. - Show the trace of the algorithm	L3	CO4	**10**

1	2	3
	4	6
7	5	8
1	2	3
4	5	6
7	8	

11.	**a.**	Explain First Order Logic inference rules for quantifier	L3	CO5	**10**
		OR			
	b.	Explain Knowledge based agent for Wumpus world agent.	L3	CO5	**10**
12.	**a.**	- The law says that it is a crime for Gaul to sell potion formulas to hostile nations. - The country Rome, an enemy of Gaul has acquired some potion formulas, and all of its formulas were sold to it by Druid Traitorix. - Traitorix is a Gaul. Prove or disprove the following using inference rules. - Is Traitorix a criminal?	L3	CO5	**10**
		OR			
	b.	Explain the properties of forward chaining and backward chaining.	L2	CO5	**10**
13.	**a.**	Demonstrate the deployment of planning system with an example.	L2	CO6	**10**
		OR			
	b.	Discuss the following planning techniques and their advantages and disadvantages. - Forward State Space Planning - Backward State Space Planning	L2	CO6	**10**